Sequences: Arithmetic & Geometric

Discovering the n-th Term through Generative Learning

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Churchill College | IGCSE Mathematics

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Session Flow: Sequences & Patterns

A 50-minute student-centered sequence.

Lesson Roadmap (50 min)

- **05 min – Phase 1:** Novelty & Curiosity (The Paper Folding Paradox)
- **10 min – Phase 2:** Expectation & Motivation (Spotting the Pattern: Add vs. Multiply)
- **20 min – Phase 3:** Effort & Scaffolding (Deducing Arithmetic & Geometric n-th Terms)
- **15 min – Phase 4:** Recognition & Peer Feedback (Sandwich Peer Review & Plenary)

1. Novelty & Curiosity | The Paper Folding Paradox

Driving Question

"If we could fold a paper sheet of thickness 0.1 mm as many times as we want...
How thick would it be after 20 folds?"

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Initial Hypotheses

Think-Pair-Share (5 min)

- Estimate the height: A few millimeters? As tall as a table? A building?
- Write down your prediction in your notebook before computing anything.

1. Novelty & Curiosity | The Paper Folding Paradox

Driving Question

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How thick would it be after 20 folds?"

Initial Hypotheses

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- Estimate the height: A few millimeters? As tall as a table? A building?
- Write down your prediction in your notebook before computing anything.

Key Inquiry

- What happens to the thickness after $1, 2, 3, 4$ folds?
- Is the thickness growing by adding a constant value or by multiplying?

2. Expectation & Motivation | Spotting the Rule

Guided Discovery (Pair Work - 5 min): Identify the growth rule for each pattern.

Sequence A

5, 8, 11, 14, 17, ...

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Guided Discovery (Pair Work - 5 min): Identify the growth rule for each pattern.

Sequence A

$$5,\ 8,\ 11,\ 14,\ 17,\ \ldots$$

Rule: Add 3 each time.

Type: Arithmetic Sequence

Sequence B

$$3,\ 6,\ 12,\ 24,\ 48,\ \ldots$$

Guided Discovery (Pair Work - 5 min): Identify the growth rule for each pattern.

Sequence A

$$5,\ 8,\ 11,\ 14,\ 17,\ \ldots$$

Rule: Add 3 each time.

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Sequence B

$$3,\ 6,\ 12,\ 24,\ 48,\ \ldots$$

Rule: Multiply by 2 each time.

Type: Geometric Sequence

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Type: Geometric Sequence

Key Definitions

- **Sequence:** An ordered, infinite list of numbers that may or may not follow a specific pattern.
- **Arithmetic Sequence:** A sequence in which the difference between consecutive terms is constant (**common difference**).
- **Geometric Sequence:** A sequence in which the ratio between consecutive terms is constant (**common ratio**).

3. How to Identify | Arithmetic vs. Geometric

Step-by-Step Test Strategy: Given a sequence $u_1, u_2, u_3, u_4, \ldots$

1. Test for Arithmetic (d)

Subtract consecutive terms:

$$u_2 - u_1, \quad u_3 - u_2, \quad u_4 - u_3$$

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Condition: If differences are equal, it's **Arithmetic**.

Common difference: $d = u_n - u_{n-1}$

2. Test for Geometric (r)

Divide consecutive terms:

$$\frac{u_2}{u_1}, \quad \frac{u_3}{u_2}, \quad \frac{u_4}{u_3}$$

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2. Test for Geometric (r)

Divide consecutive terms:

$$\frac{u_2}{u_1}, \quad \frac{u_3}{u_2}, \quad \frac{u_4}{u_3}$$

Condition: If ratios are equal, it's **Geometric**.

Common ratio: $r = \frac{u_n}{u_{n-1}}$

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Common ratio: $r = \frac{u_n}{u_{n-1}}$

Quick Summary Rule

- **Addition / Subtraction** pattern $\longrightarrow$ **Arithmetic** (Constant Difference d).
- **Multiplication / Division** pattern $\longrightarrow$ **Geometric** (Constant Ratio r).

[Level 1: Basic – Individual Work | 5 min]

Goal: Identify a (first term), d (common difference), or r (common ratio).

Task: For each sequence, specify whether it is **Arithmetic** or **Geometric**, and find a and d or r:

1. $7, 12, 17, 22, 27, \ldots$

[Level 1: Basic – Individual Work | 5 min]

Goal: Identify a (first term), d (common difference), or r (common ratio).

Task: For each sequence, specify whether it is **Arithmetic** or **Geometric**, and find a and d or r:

1. $7, 12, 17, 22, 27, \ldots$ **Arithmetic:** $a = 7, \quad d = +5$
2. $2, 6, 18, 54, 162, \ldots$

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Goal: Identify a (first term), d (common difference), or r (common ratio).

Task: For each sequence, specify whether it is **Arithmetic** or **Geometric**, and find a and d or r:

1. $7, 12, 17, 22, 27, \ldots$ **Arithmetic:** $a = 7, \quad d = +5$
2. $2, 6, 18, 54, 162, \ldots$ **Geometric:** $a = 2, \quad r = 3$
3. $80, 40, 20, 10, 5, \ldots$

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Task: For each sequence, specify whether it is **Arithmetic** or **Geometric**, and find a and d or r.

1. $7, 12, 17, 22, 27, \ldots$ **Arithmetic:** $a = 7, \quad d = +5$
2. $2, 6, 18, 54, 162, \ldots$ **Geometric:** $a = 2, \quad r = 3$
3. $80, 40, 20, 10, 5, \ldots$ **Geometric:** $a = 80, \quad r = \frac{1}{2} = 0.5$

3. Effort & Scaffolding | Level 2: Spot the Mistake!

[Level 2: Intermediate – Pair Work | 7 min]

Error Analysis Challenge: Identify the conceptual error in the student's solution.

Problem: Find the 10th term (u_{10}) of the sequence $4, 7, 10, 13, \ldots$

Student's Incorrect Solution

First term $a = 4$, Difference $d = 3$

$$u_{10} = 4 + 10 \times 3 = 4 + 30 = 34$$

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Think-Pair-Share: Where is the error?

- **Mistake:** To reach u_{10} from u_1, you only jump 9 steps, not 10!
- **Correct Formula:** $u_n = a + (n-1)d$
- **Correct Calculation:** $u_{10} = 4 + (10-1) \times 3 = 4 + 27 = 31$

[Level 3: Advanced – Group Challenge | 8 min]

Goal: Construct general n-th term formulas and solve the Paper Folding Paradox.

Arithmetic General n-th Term

$$u_n = a + (n-1)d$$

Where $a = u_1$ and $d =$ difference.

3. Effort & Scaffolding | Level 3: Deriving General Formulas

[Level 3: Advanced – Group Challenge | 8 min]

Goal: Construct general n-th term formulas and solve the Paper Folding Paradox.

Arithmetic General n-th Term

$$u_n = a + (n-1)d$$

Where $a = u_1$ and $d =$ difference.

Geometric General n-th Term

$$u_n = a \cdot r^{n-1}$$

Where $a = u_1$ and $r =$ ratio.

[Level 3: Advanced – Group Challenge | 8 min]

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Where $a = u_1$ and $d =$ difference.

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Where $a = u_1$ and $r =$ ratio.

Paper Folding Paradox Solution

- 0 folds (u_1) : 0.1 mm | 1 fold (u_2) : 0.2 mm | Geometric with $a = 0.1$, $r = 2$.
- Thickness after 20 folds (u_{21}):

$$u_{21} = 0.1 \times 2^{20} = 0.1 \times 1,048,576 = 104,857.6 \text{ mm} \approx \mathbf{104.86} \text{ meters!}$$

Notebook Exchange (8 min): Swap notebooks with a neighboring group.

Verification Criteria (What to check)

1. **Sequence Type:** Did they correctly distinguish Arithmetic vs. Geometric?
2. **Formula Setup:** Did they use $(n-1)$ instead of n?
3. **Calculation Accuracy:** Is the paper folding height computed correctly?

Notebook Exchange (8 min): Swap notebooks with a neighboring group.

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Layer 1 (Top Bread): **Specific Praise** (Highlight a clear strength).

Layer 2 (Filling): **Constructive Check** (Ask a technical question to fix an error).

Layer 3 (Bottom Bread): **Encouragement & Signature** (Sign at the bottom).

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule		

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
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n-th Term Formula		

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
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Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r
n-th Term Formula	$u_n = a + (n-1)d$	$u_n = a \cdot r^{n-1}$

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r
n-th Term Formula	$u_n = a + (n-1)d$	$u_n = a \cdot r^{n-1}$
Growth Behavior		

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r
n-th Term Formula	$u_n = a + (n-1)d$	$u_n = a \cdot r^{n-1}$
Growth Behavior	Linear growth	

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Example		

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Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r
n-th Term Formula	$u_n = a + (n-1)d$	$u_n = a \cdot r^{n-1}$
Growth Behavior	Linear growth	Exponential growth
Example	$3, 7, 11, 15 \quad (d = 4)$	

Plenary & Session Synthesis

Concept	Arithmetic Sequence	Geometric Sequence
Pattern Rule	Add/subtract d	Multiply/divide r
n-th Term Formula	$u_n = a + (n-1)d$	$u_n = a \cdot r^{n-1}$
Growth Behavior	Linear growth	Exponential growth
Example	$3, 7, 11, 15 \quad (d = 4)$	$3, 6, 12, 24 \quad (r = 2)$

Great job mastering sequences and patterns today!

Sequences Practice Workshop

Arithmetic & Geometric Sequences: Problem Solving Session

M. Sc. Guillermo Gachuz Atitlán

October 20, 2026

Churchill College | IGCSE Mathematics

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Session Flow: Sequences Practice Workshop

A 50-minute generative problem-solving sequence (PGC Framework).

Lesson Roadmap (50 min)

- **05 min – Phase 1:** Novelty & Setup (Quick Strategy Recall & Challenge Launch)
- **15 min – Phase 2:** Solo Practice (Independent Student Problem Solving)
- **20 min – Phase 3:** Guided Resolution & Teacher Circulation (Addressing Doubts & Board Solutions)
- **10 min – Phase 4:** Reflection & Key Insights (Shared Hints & Common Pitfalls)

1. Setup & Instructions | Individual Challenge

Workshop Goal

"Analyze each sequence, justify its type (Arithmetic or Geometric), and solve for the required term or index n."

15-Min Solo Work Rules

- Work **independently** in your notebook.
- Show explicit justification: calculate $d = u_2 - u_1$ or $r = \frac{u_2}{u_1}$.
- Write down the general u_n formula before substituting values.

Teacher Circulation

- Raise your hand if you get stuck on solving for n using logarithms.
- Focus on identifying whether you add a constant (d) or multiply by a factor (r).

2. Practice Problems | Set A (Problems 1–3)

Problem 1

Consider the sequence:

$$\frac{1}{2},\ 1,\ \frac{3}{2},\ 2,\ \ldots$$

a) Identify the type of sequence (justify).

b) Find the n-th term (u_n).

c) Find the term u_{30}.

Problem 2

Consider the sequence:

$$5,\ 10,\ 20,\ 40,\ \ldots$$

a) Identify the type of sequence (justify).

b) Find the n-th term (u_n).

c) Find the term u_{20}.

Problem 3

Consider the sequence:

$$10,\ 5,\ 2.5,\ \ldots$$

a) Identify the type of sequence (justify).

b) Find the n-th term (u_n).

c) Find the term u_{10}.

Solo Practice: 7 Minutes

2. Practice Problems | Set B (Problems 4–6)

Problem 4

Consider the sequence:

$$2,\ 6,\ 10,\ 14,\ \ldots$$

a) Identify the type of sequence (justify).

b) If $u_n = 122$, find n.

Problem 5

Consider the sequence:

$$9,\ 3,\ 1,\ \ldots$$

a) Identify the type of sequence (justify).

b) If $u_n = \frac{1}{2187}$, find n.

Problem 6

Consider the sequence:

$$2,\ 8,\ 32,\ \ldots$$

a) Identify the type of sequence (justify).

b) If $u_n = 134,217,728$, find n.

Solo Practice: 8 Minutes

3. Step-by-Step Solutions | Problems 1–3

Solution 1

a) Type: Arithmetic.
Justification: Constant difference $d = 1 - \frac{1}{2} = \frac{1}{2}$.

b) n-th term:

$$u_n = a + (n-1)d$$

$$u_n = \frac{1}{2} + (n-1)\frac{1}{2} = \boxed{\frac{n}{2}}$$

c) Find u_{30}:

$$u_{30} = \frac{30}{2} = \boxed{15}$$

Solution 2

a) Type: Geometric.
Justification: Constant ratio $r = \frac{10}{5} = 2$.

b) n-th term:

$$u_n = a \cdot r^{n-1}$$

$$u_n = \boxed{5 \cdot 2^{n-1}}$$

c) Find u_{20}:

$$u_{20} = 5 \cdot 2^{19} = 5 \cdot 524,288$$

$$u_{20} = \boxed{2,621,440}$$

Solution 3

a) Type: Geometric.
Justification: Constant ratio $r = \frac{5}{10} = \frac{1}{2}$.

b) n-th term:

$$u_n = 10 \cdot \left(\frac{1}{2}\right)^{n-1}$$

c) Find u_{10}:

$$u_{10} = 10 \cdot \left(\frac{1}{2}\right)^{9} = \frac{10}{512} = \boxed{\frac{5}{256}}$$

3. Step-by-Step Solutions | Problems 4–6

Solution 4

a) Type: Arithmetic.
Justification: $d = 6 - 2 = 4$.

$$u_n = 2 + (n-1)4 = \mathbf{4n - 2}$$

b) Find n if $u_n = 122$:

$$4n - 2 = 122$$

$$4n = 124 \implies \mathbf{n = 31}$$

Solution 5

a) Type: Geometric.
Justification: $r = \frac{3}{9} = \frac{1}{3}$.

$$u_n = 9 \cdot \left(\frac{1}{3}\right)^{n-1}$$

b) Find n using $\log_{\frac{1}{3}}$:

$$9 \cdot \left(\frac{1}{3}\right)^{n-1} = \frac{1}{2187}$$

$$\left(\frac{1}{3}\right)^{n-1} = \frac{1}{19683}$$

Use $\log_{\frac{1}{3}}$:

$$n - 1 = \log_{\frac{1}{3}}\left(\frac{1}{19683}\right) \implies n - 1 = 9$$

Therefore, $n = 10$.

Solution 6

a) Type: Geometric.
Justification: $r = \frac{8}{2} = 4$.

$$u_n = 2 \cdot 4^{n-1}$$

b) Find n using $\log_4$:

$$2 \cdot 4^{n-1} = 134,217,728$$

$$4^{n-1} = \frac{134,217,728}{2} = 67,108,864$$

Use $\log_4$:

$$n - 1 = \log_4(67,108,864) = 13$$

Therefore, $n = 14$.

4. Synthesis & Shared Key Hints

Key Strategy Takeaways

Logarithmic properties used to solve exponential equations:

1. Base-b Logarithm Identity

Using the logarithm with matching base eliminates the exponent via the power rule:

$$\log_b(b^k) = k \cdot \log_b(b) = k$$

This directly isolates linear expressions in n.

2. Step-by-Step Procedure

1. Isolate the exponential factor ($b^k = C$).
2. Apply $\log_b$ to both sides.
3. Solve the resulting linear equation for n.

Great job mastering sequence problem solving today!

Problem Solving: Arithmetic & Geometric Sequences

Generative Practice & Peer Learning Session

M. Sc. Guillermo Gachuz Atitlán

October 20, 2026

Churchill College | IGCSE Mathematics

MSc Gachuz Atitlán

Session Flow: Problem Solving Workshop

A 50-minute generative problem-solving workshop.

Lesson Roadmap (50 min)

- **20 min – Phase 1: Individual Effort & Question Formulation**
  - Silent focus: solve exercises independently.
  - If stuck, construct a **meaningful "Helper Question"** that, if answered, would give you the missing step (e.g., *"How do I find r when the sequence decreases?"*). **No generic "I don't know" / "I'm lost" allowed!**
- **15 min – Phase 2: Peer Peer-Consultation**
  - Pair up, ask your helper questions, and resolve remaining gaps together.
- **10 min – Phase 3: Cold-Call Sharing & Problem Debrief**
  - Randomly selected students present key insights or tricky steps.
- **05 min – Phase 4: Metacognitive Closure (Exit Reflection)**
  - Reflection on breakthrough questions and common pitfalls.

Workshop Protocol: 20 – 15 – 10 – 5

1. Solo Focus (20 min)

- Work on the problem sheet independently.
- **Stuck Rule:** Write down: *"What exact question do I need to ask to unlock this problem?"*
- *Example:* "How do I find r when terms are negative?"

2. Peer Consultation (15 min)

- Pair with a neighbor.
- Read your **Helper Questions** to each other first.
- Help each other solve uncompleted tasks.

3. Random Sharing (10 min)

- Random cold-calling for problem solutions.
- Be ready to explain **how** your pair solved it or what question unlocked it.

4. Metacognitive Closure (5 min)

- Class debrief on key conceptual traps (n vs. $n-1$, common ratio vs. difference).

Session Closure: Metacognition & Takeaways

Exit Reflection: The Power of Good Questions

Before handing in your work, write down at the top of your page:

1. **Key Breakthrough:** What was the most useful question asked during pair work today?
2. **Top Pitfall to Avoid:** What is one detail you must double-check in future exam questions?

Session Closure: Metacognition & Takeaways

Exit Reflection: The Power of Good Questions

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Arithmetic Checklist

- Constant difference $d = u_n - u_{n-1}$
- $u_n = a + (n-1)d$

Geometric Checklist

- Constant ratio $r = \frac{u_n}{u_{n-1}}$
- $u_n = a \cdot r^{n-1}$

Great effort on autonomous problem solving today!

Advanced Sequences & Fractals

Simultaneous Equations, Algebraic Patterns & Fractal Geometry

M. Sc. Guillermo Gachuz Atitlán

October 20, 2026

Churchill College | IGCSE / IB Mathematics

MSc Gachuz Atitlán

Session Flow: Advanced Sequences & Fractals

A 50-minute generative problem-solving session on systems of equations and fractals.

Lesson Roadmap (50 min)

- **05 min – Phase 1:** Setup & Systems Recall (Simultaneous Equations Strategies)
- **15 min – Phase 2:** Solo Practice (Independent Problem Solving)
- **20 min – Phase 3:** Guided Resolution & Teacher Circulation
- **10 min – Phase 4:** Reflection & Key Insights (Fractal Growth Rules & Common Pitfalls)

M. Sc. Guillermo Gachuz Atitlán

1. Setup & Instructions | Advanced Challenge

Workshop Goal

"Set up systems of two equations to solve for u_n, d, or r, and apply geometric patterns to recursive fractal models."

Algebraic Systems Strategy

When given a **system of two equations with two variables**:

1. **Choose & Isolate:** Select one equation and solve for one of the variables.
2. **Substitute:** Substitute this isolated expression into the second equation.
3. **Solve:** Solve the resulting single-variable equation.
4. **Back-Substitute:** Substitute the known value back into the first equation to find the second variable.

Fractal Modeling Strategy

- Identify u_1 or base value.
- Determine the scaling multiplier r (ratio) between Step n and Step $n+1$.
- Build the geometric formula: $u_n = a \cdot r^n - 1$.

Quadratic Formula

Remember that to solve an equation of the form $ax^2 + bx + c = 0$, we use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Practice Problems | Set A: Algebraic Systems

1 Arithmetic System

An arithmetic sequence has $u_4 = 17$ and $u_9 = 42$.

a) Form 2 equations and solve for a and d.

b) Find n if $u_n = 102$.

2 Geometric System

A geometric sequence has $u_3 = 12$ and $u_6 = 96$.

a) Form 2 equations and solve for a and r.

b) Calculate u_{10}.

3 Algebraic Terms

The terms $k - 1$, $2k + 1$, and $5k + 3$ form a geometric sequence.

a) Show that $k^2 - 6k - 4 = 0$.

b) Assuming $k > 0$, find the exact ratio r.

Solo Practice: 7 Minutes

4 Sierpinski

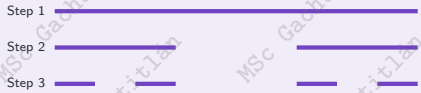


A Sierpinski Triangle starts with 1 solid purple equilateral triangle at Step 1. At each step, the central inverted triangle is removed, creating 3 smaller black triangles.

1. Write down the number of black triangles N_n for Steps 1, 2, 3, and 4.
2. State the n-th term formula for N_n and calculate the number of black triangles at Step 8.

Solo Practice: 8 Minutes

5 Cantor Set



The Cantor Set starts at Step 1 with a line segment of length $L_1 = 1$ m. At each step, the open middle third of every remaining interval is removed.

1. Find the total remaining length L_n after n steps.
2. Determine after how many steps n the total length first falls below 0.01 m.

Solo Practice: 8 Minutes

6 Koch Snowflake



Step 1



Step 2

The Koch Snowflake starts at Step 1 with an equilateral triangle of perimeter $P_1 = 3$ cm. At each iteration, the middle third of every segment is replaced by two segments of equal length.

1. Draw Step 3.
2. Express the perimeter P_n after n iterations as a geometric sequence.
3. Calculate the total perimeter P_5 correct to 2 decimal places.

Solo Practice: 8 Minutes

Solution 1 (Substitution)

a) System of Equations:

$$a + 3d = 17 \quad (1)$$
$$a + 8d = 42 \quad (2)$$

1. Isolate a in (1):

$$a = 17 - 3d$$

2. Substitute a into (2):

$$(17 - 3d) + 8d = 42$$

3. Solve for d:

$$17 + 5d = 42 \implies 5d = 25 \implies \mathbf{d = 5}$$

4. Back-substitute d:

$$a = 17 - 3(5) \implies \mathbf{a = 2}$$

b) Find n for $u_n = 102$:

$$2 + (n-1)5 = 102 \implies 5n - 3 = 102$$
$$5n = 105 \implies \mathbf{n = 21}$$

Solution 2 (Substitution)

a) System of Equations:

$$a \cdot r^2 = 12 \quad (1)$$
$$a \cdot r^5 = 96 \quad (2)$$

1. Isolate a in (1):

$$a = \frac{12}{r^2}$$

2. Substitute a into (2):

$$\left(\frac{12}{r^2}\right) \cdot r^5 = 96$$

3. Solve for r:

$$12r^3 = 96 \implies r^3 = 8 \implies \mathbf{r = 2}$$

4. Back-substitute r:

$$a = \frac{12}{2^2} = \frac{12}{4} \implies \mathbf{a = 3}$$

b) Calculate u_{10}:

$$u_{10} = 3 \cdot 2^9 = 3 \cdot 512 = \mathbf{1536}$$

Solution 3

a) Ratio Property: $\frac{u_2}{u_1} = \frac{u_3}{u_2}$

$$\frac{2k+1}{k-1} = \frac{5k+3}{2k+1}$$

$$(2k+1)^2 = (k-1)(5k+3)$$

$$4k^2 + 4k + 1 = 5k^2 - 2k - 3$$

$$\mathbf{k^2 - 6k - 4 = 0}$$

b) Solve for $k > 0$ and r:

$$k = \frac{6 + \sqrt{52}}{2} = 3 + \sqrt{13}$$

$$r = \frac{2(3+\sqrt{13})+1}{(3+\sqrt{13})-1} = \mathbf{1 + \sqrt{13}}$$

3. Step-by-Step Solutions | Problems 4–6

Solution 4 (Sierpinski)

a) Triangle count N_n:

$$\text{Step 1 } (n=1): a = 1$$

$$\text{Step 2 } (n=2): N_2 = 3 \implies r = 3$$

Formula $N_n = a \cdot r^{n-1}$:

$$N_n = 1 \cdot 3^{n-1} = \mathbf{3^{n-1}}$$

Values for Steps 1, 2, 3, 4:

$$3^0 = 1,\ 3^1 = 3,\ 3^2 = 9,\ 3^3 = 27$$

b) Calculate Step 8 ($n = 8$):

$$N_8 = 3^{8-1} = 3^7 = \mathbf{2187} \text{ triangles}$$

Solution 5 (Cantor Set)

a) Length formula L_n:

$$\text{Step 2 } (n=2): L_2 = \frac{2}{3} \implies r = \frac{2}{3}$$

Formula $L_n = a \cdot r^{n-1}$:

$$L_n = 1 \cdot \left(\frac{2}{3}\right)^{n-1} = \mathbf{\left(\frac{2}{3}\right)^{n-1}}$$

b) Solve for $L_n < 0.01$:

$$\left(\frac{2}{3}\right)^{n-1} < 0.01$$

Apply $\log_{\frac{2}{3}}$ (flips inequality!):

$$n - 1 > \log_{\frac{2}{3}}(0.01) \approx 11.35$$

$$n > 12.35 \implies \mathbf{n = 13} \text{ steps}$$

Solution 6 (Koch Snowflake)

a) Perimeter formula P_n:

$$\text{Step 1 } (n=1): a = 3 \text{ cm}$$

Each side split gives ratio $r = \frac{4}{3}$. Formula $P_n = a \cdot r^{n-1}$:

$$P_n = \mathbf{3} \cdot \left(\frac{4}{3}\right)^{\mathbf{n-1}}$$

b) Calculate P_5 ($n = 5$):

$$P_5 = 3 \cdot \left(\frac{4}{3}\right)^{5-1} = 3 \cdot \left(\frac{4}{3}\right)^4$$

$$P_5 = 3 \cdot \frac{256}{81} = \frac{256}{27} \approx \mathbf{9.48} \text{ cm}$$

4. Synthesis & Shared Key Hints

Key Strategy Takeaways

Summary of algebraic and geometric properties applied today:

1. Elimination & Substitution

- Substitution works universally: isolate one variable ($a = f(d)$ or $a = f(r)$) and replace it into the second equation.

2. Fractals as Sequences

- Count, length, and perimeter change exponentially at each step n.
- Watch out for inequality flips when taking logarithms with bases < 1 (e.g., $\log_{2/3}$).

Great job mastering advanced sequences today!